

Title: Dimension, Measure, Metric — A Unified View from Cosmic Expansion to Computer Systems

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ABSTRACT

Today we started with an abstract question: What is measure? What is dimension? What is metric?

Along the way, we touched on:

- Cosmic expansion and Hubble's law
- Riemannian geometry and General Relativity
- The constancy of light speed and the truth behind "superluminal recession"
- Kolmogorov's axioms in probability spaces
- A plain-language understanding of zero measure
- Computer systems: NIC bandwidth, RDMA, early termination in HNSW

This blog is not about showing off跨disciplinary knowledge.

It's about showing a fact: the three-layer structure of

"type of unit" → "how many units" → "how large a unit actually is"

appears repeatedly in physics, mathematics, and computer science.

PART 1: THE THREE-LAYER FRAMEWORK — DIMENSION · MEASURE · METRIC

1.1 Dimension

What it asks: What "type" of quantity are we measuring?

Examples:

- Length → dimension [L]
- Time → dimension [T]
- Mass → dimension [M]
- Probability → dimensionless, but can be treated as a logical category of "possibility"

Dimension does not answer "how much", only "what type".

1.2 Measure

What it asks: How many units?

Examples:

- Lebesgue measure of interval $[0,1] = 1$ (a pure number, not 1 meter)
- Measure of a single point = 0
- Measure of finitely many points = 0

- Measure of all rational numbers = 0 (densely distributed but still zero measure)

Plain-language understanding of zero measure:

In terms of "volume/length/probability", a zero-measure set is negligibly small, as if it doesn't exist.

But "measure zero" does NOT mean "impossible".

For example, picking exactly 0.5 from $[0,1]$ has probability 0, yet it is possible.

1.3 Metric / Metric Tensor

What it asks: Here and now, in this direction, how much physical quantity does 1 unit correspond to?

Fixed metric (Euclidean geometry):

- 1 unit = 1 meter, forever fixed
- Distance = coordinate difference $\times$ 1

Dynamic metric (cosmic expansion, FLRW):

- 1 unit = $a(t)$ meters, where $a(t)$ grows over time
- Comoving coordinate difference (how many units) stays constant
- Physical distance = comoving difference $\times a(t)$

Curved metric (Riemannian geometry / General Relativity):

- The size of 1 unit can vary with position, direction, and time
- Described by the metric tensor $g_{\mu\nu}$

Key insight:

The "number of grid cells" (measure) in comoving coordinates stays constant, but the "physical size of each cell" (the metric's $a(t)$) grows over time.

→ This is the mathematical essence of cosmic expansion: galaxies recede without moving.

PART 2: COSMIC EXPANSION · RIEMANNIAN GEOMETRY · CONSTANCY OF LIGHT SPEED

2.1 Why "superluminal recession" does NOT violate relativity

Locally, in any inertial frame: light speed is still c

What exceeds c is the "coordinate velocity" — an apparent speed computed in comoving coordinates

Physical information, matter, and causality can never exceed c

Conclusion:

The constancy of light speed is a statement about the "local metric", not about any global coordinate system.

2.2 Riemannian vs Euclidean geometry

Euclidean geometry: a single fixed ruler, invariant everywhere, every direction, every time
Riemannian geometry: the ruler can vary with position, direction, and time

Core idea of General Relativity:

Matter and energy tell spacetime how to curve (how the metric changes)

The curvature of spacetime tells matter how to move

2.3 Mathematical expression of cosmic expansion

FLRW metric:

$$ds^2 = -c^2 dt^2 + a(t)^2 [dx^2 + dy^2 + dz^2]$$

$a(t)$: scale factor

- $a(t) = 1$ (fixed) $\rightarrow$ ordinary Euclidean space

- $a(t)$ increases with time $\rightarrow$ units grow

- Galaxies have fixed comoving coordinates $\rightarrow$ physical distance $\propto a(t)$

Redshift explained:

Light waves are stretched by expanding space; wavelength $\propto 1/a(t)$

PART 3: ANALOGY IN PROBABILITY SPACE (KOLMOGOROV'S AXIOMS)

3.1 Mapping the three-layer framework to probability theory

Dimension $\rightarrow$ "Probability" (what we are measuring is likelihood)

Measure $\rightarrow$ Probability value P (a pure number between 0 and 1)

Metric $\rightarrow$ Total measure = 1 (the global normalization condition)

Explanation:

In Kolmogorov's axioms, the total probability of the entire sample space must be 1.

This "total measure = 1" acts as the "metric" of probability space — it anchors "1 probability unit" to "certainty".

3.2 Zero measure in probability

In a continuous distribution, the probability of any single specific point = 0

But the probability of an interval, say $[0, 0.5]$, can be 0.5

Hence: "probability 0" does NOT equal "impossible"

PART 4: MAPPING TO COMPUTER SYSTEMS — BANDWIDTH · RDMA · HNSW

4.1 Dimension, measure, metric in computing

Dimension $\rightarrow$ type of physical quantity (bytes, seconds, requests)

Measure $\rightarrow$ quantity (10 GB, 1000 QPS, 100 billion log entries)

Metric $\rightarrow$ actual size of unit / conversion rule

- NIC: 1 Gb/s = 125 MB/s (actual $\approx$ 100 MB/s)
- Memory bandwidth: DDR4 $\approx$ 25 GB/s
- SSD: SATA $\approx$ 500 MB/s, NVMe $\approx$ 3 GB/s
- HDD: $\approx$ 150 MB/s

4.2 RDMA — the path that bypasses main memory

Traditional path: NIC $\rightarrow$ main memory $\rightarrow$ CPU $\rightarrow$ main memory $\rightarrow$ SSD

RDMA path: NIC $\rightarrow$ SSD directly (bypasses main memory)

Benefits:

- Zero copy
- CPU offload
- 30-50% lower latency
- 100GbE RDMA 实测 throughput $\approx$ 11.6 GB/s (write) / 12.4 GB/s (read)

4.3 HNSW and early termination

Problem:

In distributed HNSW (1000 machines), how to make most machines stop early?

Answer:

Use spatial partitioning (not random sharding)

$\rightarrow$ Each query hits only 1-5 machines

$\rightarrow$ Other machines discover at the top-level entry point that the distance is already too large, and stop immediately

$\rightarrow$ Cost: tens of microseconds vs milliseconds

Early termination condition:

Inner product similarity $>$ threshold T (e.g., 0.01)

If current best similarity already exceeds T , return early without traversing deeper.

Impact on throughput:

- Single-machine early termination $\rightarrow$ QPS increases 2-5x (recall loss $<$ 5%)
- Distributed + routing layer filtering $\rightarrow$ total throughput scales linearly with number of machines

PART 5: KEY CONCLUSIONS (English Summary)

1. Dimension asks "what type"
2. Measure asks "how many" (a pure number)
3. Metric asks "right here, right now, how large is 1 unit"
4. Cosmic expansion = the unit of the metric grows, while "how many units" (comoving coordinate difference) stays fixed

5. Constancy of light speed = always true in the "local metric"; does not prevent coordinate velocity from exceeding c
6. Riemannian geometry = allows the metric to vary with position, direction, and time
7. Probability theory = total measure = 1 is its "metric", anchoring the unit of probability
8. Zero measure = in the sense of measurement, it is negligible, but not impossible
9. Computer system bottleneck = the slowest link determines overall performance (NIC/HDD/SSD/memory/CPU)
10. Optimal distributed retrieval = most machines do shallow filtering (tens of microseconds), few machines do deep search

PART 6: TERMINOLOGY TABLE

Chinese	English
量纲	Dimension
测度	Measure
度规	Metric / Metric Tensor
共动坐标	Comoving Coordinates
物理坐标	Physical Coordinates
尺度因子	Scale Factor $a(t)$
零测度	Zero Measure / Null Set
几乎处处	Almost Everywhere
黎曼几何	Riemannian Geometry
欧氏几何	Euclidean Geometry
广义相对论	General Relativity
FLRW 度规	FLRW Metric
光速不变	Constancy of Light Speed
RDMA	Remote Direct Memory Access
提前终止	Early Termination
HNSW	Hierarchical Navigable Small World
QPS	Queries Per Second

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